



Application of real-time surgical navigation for zygomatic fracture reduction and fixation

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Received 29 January 2020; accepted 27 May 2021

Available online xxx

KEYWORDS

Computer-assisted navigation system;
Real-time surgical navigation;
Zygomatic fracture;
Zygomaticomaxillary complex

Summary Background: Inappropriate treatment of zygomatic fractures can reduce esthetic and functional outcomes. The aim of this study was to answer the research question: "Among patients with a unilateral zygomatic fracture, is the use of computer-assisted real-time navigation system during fracture reduction precise and accurate to create postoperative facial symmetry?"

Methods: Using a retrospective cohort study design, we enrolled a cohort of unilateral zygomatic fractures undergoing open reduction and internal fixation (ORIF) with the aid of the computer-based navigation system at Chang Gung Memorial Hospital, Taiwan, during January 2015 and March 2018. The predictor variable was the comparison before and after surgery. The main outcome variables included (1) two-dimensional (2D) reduction of the displacement at five anatomical landmarks: zygomaticofrontal, inferior orbital rim, zygomaticosphenoidal, zygomaticomaxillary, and zygomaticotemporal lines/butresses and (2) three-dimensional (3D) differences on distances between zygomatic surface to the porion plane and the midpoint of zygomatic arch (ZA) to the mid-porion (MP) plane. The Wilcoxon signed-rank test was computed to compare between pre- and postoperative data, and a p-value less than 0.05 was considered statistically significant.

Results: The cohort comprised 24 subjects (50% females, 75% left-sided) with a mean age of 30.5 ± 13.8 years. On 2D analysis, the significant fracture reduction was found: 4.78 vs. 1.22 mm, 1.78 vs. 0.40 mm, 3.50 vs. 0.07 mm, 3.06 vs. 0.55 mm, and 2.55 vs. 0.50 mm at

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zygomaticomaxillary, zygomaticofrontal, inferior orbital rim, zygomaticosphenoidal, and zygomaticotemporal landmarks. The 3D evaluations revealed the significant reduction of the differences between the left and right zygomatic surface to the porion plane (4.09 ± 2.12 vs. 0.46 ± 0.35 mm) and between the left and right ZA midpoints to the MP plane (4.89 ± 2.59 vs. 0.71 ± 0.44 mm) ($p < 0.001$ for both 2D and 3D analyses).

Conclusions: The results of this study suggest that the real-time surgical navigation system can effectively guide the ORIF of zygomatic fractures. Future research studies should focus on the learning curve and cost-effectiveness analysis of this technique.

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Introduction

Zygomaticomaxillary complex (ZMC) fracture accounts for about 25% of facial fractures, and it is viewed as the second most common facial fracture pattern following nasal fractures^{1,2}. It could result from traffic accidents, a direct punch, or falling down. Linear fracture(s) without displacement can be treated conservatively, while surgical reduction is indicated in a displaced or dislocated fracture³. Fractures of the ZMC or zygomatic arch (ZA) often lead to asymmetrical facial appearance and functional impairment⁴. Inadequate reduction of fractures leads to poor esthetic outcomes and further complications⁵. However, accurate reduction in ZMC fractures, particularly for severely comminuted type, is a challenge for plastic or oral and maxillofacial surgeons⁶. Conventionally, the zygoma fracture reduction was confirmed by tactile and direct visualization during operation. However, this would be difficult due to the swelling of the soft tissue that hampered clinical evaluation⁷. To improve surgical results, several techniques were innovated. These tools optimize operative planning and reduce complications⁸.

Among these techniques, the real-time surgical navigation system has been widely used and became increasingly popular in craniofacial surgery for congenital malformation or trauma reconstruction⁹. The advantages are better precision, accuracy, effectiveness, predictability, and improvement of the symmetry in unilateral or bilateral ZMC fractures⁷. The optimal 3D virtual surgical plans can be obtained preoperatively through computer simulation by image mirroring. During operation, the fractured bone segments were reduced and their position can be checked by real-time tracking frequently through the navigation pointer based on the navigation system. An adequate reduction can be ensured intraoperatively to match the preoperative designed location of the fractured zygoma, to achieve more esthetic results¹⁰.

To demonstrate the potential of intraoperative navigation in the management of craniofacial trauma, we have selected unilateral zygomatic fractures to provide the opportunity to assess the outcomes to the uninjured contralateral side. As our understanding of the benefits and limitations of this approach are better delineated, we can better appreciate the best utilization of this technology. Our initial experience would suggest that navigation will be more beneficial with injuries that produce greater displacement and comminution. Hence, the aim of this study was to answer the clinical research question: "Among patients with a unilateral zygomatic fracture, is the use of computer-assisted

real-time navigation system during fracture reduction precise and accurate to create postoperative facial symmetry?" We hypothesized that navigation-assisted zygomatic reduction would significantly reduce facial asymmetry. Our specific aims were to (1) conduct a retrospective study identifying unilateral zygomatic fractures in our hospital, (2) measure the differences of relevant anatomical landmarks comparing before and after the surgery, and (3) provide our surgical technique on unilateral zygomatic reduction and internal fixation using the intraoperative navigation system.

Materials and methods

The predictor variable was the comparison before and after surgery. The main outcome variables included (1) two-dimensional (2D) reduction of the displacement at five anatomical landmarks: zygomaticofrontal, inferior orbital rim, zygomaticosphenoidal, zygomaticomaxillary, and zygomaticotemporal lines/butresses and (2) three-dimensional (3D) differences in distances between zygomatic surface to the porion plane and midpoint of ZA to the mid-porion (MP) plane.

Patients

This study was approved by the institutional review board of Chang Gung Memorial Hospital, Linkou Branch, Taiwan (IRB: 201900292B0). Informed consent was obtained from the patients. The inclusion criteria were acute unilateral zygomatic fractures that resulted from traffic accidents, falls, or human altercations. Patients with zygomatic fractures more than 1 month would be excluded. Preoperative and postoperative signs and symptoms were recorded. Fractures were reduced and fixed by the guidance of the 3D computer-assisted navigation system.

Materials

Preoperative facial bone computed tomography (CT) imaging was obtained one day before surgery with 1-mm collimation/slice thickness for every patient. Preoperative virtual planning and simulation were performed with BrainLAB 3D modeling software iPlan CMF (BrainLAB AG, Munich, Germany). Digital patient data were uploaded and processed with Digital Imaging and Communications in Medicine (DICOM), which included subsequent object creation, mir-

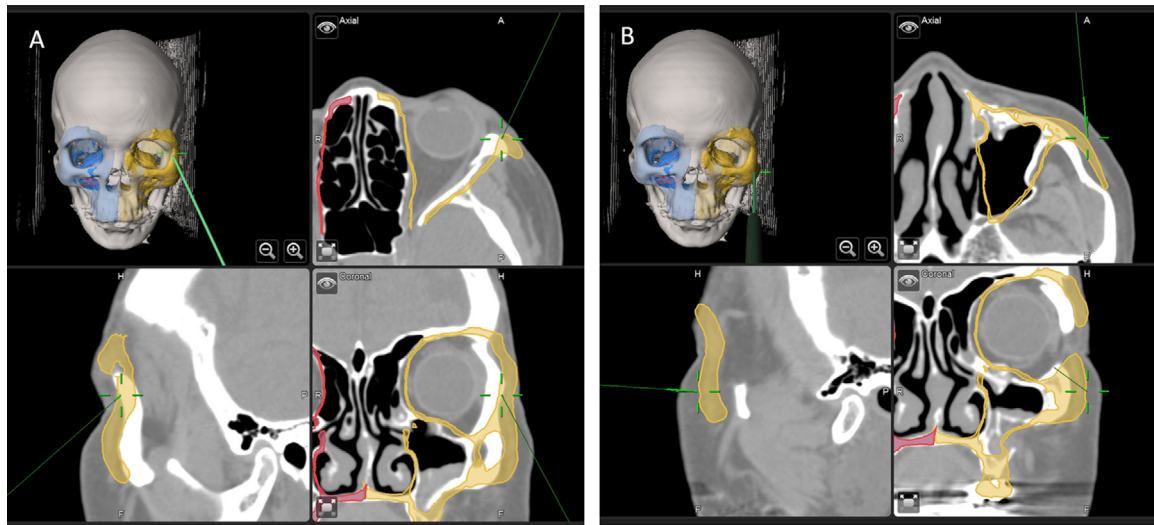


Figure 1 Computed tomography (CT) image of left zygoma fracture displayed by navigation system (BrainLAB). The design of ideal reduction position is shown in yellow outline. The green probe indicates real-time bone position before (A) and after reduction (B).

roring the unaffected side of the zygomatic bone to the affected side, objective data measurement, and data export (Figure 1). The BrainLAB Vector Vision navigation platform was used for the confirmation of symmetrical zygoma reduction during surgery. As part of the preoperative preparation, a skull reference tool adapter with three reflective glass-grain coating plastic spheres (Latero Reference Star, BrainLAB) (Figure 2) was secured to the contralateral parietal bone with one self-tapping titanium screw without hair shaving. Surface matching was conducted with a laser pointer (Z-touch BrainLAB) by scanning the periorbital and frontal skin. Upper central incisors were also used as anatomical landmarks when verifying referencing accuracy by placing the tip of the navigation pointer at the surface of the upper central incisor and the buccal embrasure of the upper central incisors to verify the precision in the X-, Y-, and Z-axis. Deviation between the preoperative CT image and pointer landmark should be < 1 mm; thereafter, successful registration was confirmed¹¹.

Surgical procedures

Surgery was performed by a single surgeon. Peri-operative antibiotics of cefazolin and clindamycin were administered 30 min before surgery. Under general anesthesia, the patient was placed in a supine position. The scalp, face, and neck were prepped and draped in a sterile fashion. The oral cavity was irrigated with diluted povidone-iodine solution. Surgical incisions were made onto the ipsilateral upper gingivobuccal sulcus mucosa. Additional incisions, including transconjunctival, subciliary, bicoronal, or preauricular cutting lines, were made depending on fracture pattern and/or difficulty in reduction. Soft tissue was carefully degloved from the bone, and the infraorbital nerve (ION) was well preserved. Adequate reduction was performed under the guidance of a real-time surgical navigation system. Each suture line and critical point were repeatedly examined by the pointer to ensure precise reduction, matched

up to preoperative planning image boundary. Fixations were made by titanium or resorbable screws and plates. One to four fracture points were fixed, which depends on bony stability. After normal saline irrigation of the surgical field, all incisions were closed in layers. The skull tool adapter and self-tapping screw were removed from the skull. No drain was placed intraorally. Occasionally, a Penrose drain was placed onto the orbital floor for the prevention of retrobulbar hematoma when a subciliary incision was made for lateral orbital wall reduction or a concomitant orbital floor reconstruction.

Cefazolin and clindamycin were administered postoperatively for 3 days. An ice pack was applied on the face for edema control. A soft food diet was suggested for the prevention of the bone refracture or screw and plate loosening when biting or chewing, particularly for patients who had the fracture involving zygomaticotemporal suture.

Two-dimensional (2D) radiological assessment

Postoperative CT was arranged 6-12 months later, with an average of 298 days. The outcome was evaluated by the severity of displacement at each suture line (Zygomaticofrontal: ZF, Inferior orbital rim: IO, Zygomaticosphenoidal: ZS, Zygomaticomaxillary: ZM, and Zygomaticotemporal: ZT) using the facial bone CT for 2D assessment, preoperative, and postoperatively as in our previous report¹². We measured the degree of displacement in millimeters at these five key suture lines, and the average sum of displacement was compared between preoperative and postoperative data. Bone displacement < 2 mm indicated a fair surgical result because it was clinically unrecognizable.

3D image assessment

Pre- and postoperative CT 3D images were assessed by Simplant O&O 11.0. Frankfort Horizontal plane (FH) is a stan-



Figure 2 Preoperative setting. A skull reference tool adapter with three reflective glass-grain coating plastic spheres was secured to the contralateral parietal bone without hair shaving.

dard reference plane passing through the bilateral porions and the left orbital. From CT images, the left porion and right porion were identified. The midpoint between two porions was marked PoMid. A coronal plane passing through both porions, perpendicular to the FH, was defined as the porion (P) plane. A sagittal plane passing through the PoMid, perpendicular to the FH, was defined as the MP plane. The left midpoint of zygoma arch (mid-ZAL) and right midpoint of zygoma arch (mid-ZAR) were also identified. The distances between mid-ZAL to MP plane (D-ZAL) and mid-ZAR to MP plane (D-ZAR) were measured pre- (Figure 3A) and postoperatively (Figure 3B). This distance indicated the status of protrusion of the ZA. Also, the zygomatic surface point (ZSL or ZSR) was defined as the most anteriorly protruded point of the zygoma body. Distances between ZSL to P plane (D-ZSL) and ZSR to P plane (D-ZSR) were also measured pre- (Figure 3A) and postoperatively (Figure 3B). This distance indicated the condition of anterior protrusion of the zygoma body. Pre- and postoperative differences between D-ZAL vs. D-ZAR and D-ZSL vs. D-ZSR were calculated.

Statistics

Demographic data were presented as frequencies or mean $\pm$ standard deviation. The Wilcoxon signed-rank test was used to compare pre- and postoperative measurement data of 2D

radiology and 3D image assessment. P value <0.05 indicated statistical significance.

Postoperative complications

Postoperative complications were documented. They were categorized into two groups: (1) transient effects, which were observed within 2 weeks after surgery and (2) chronic sequelae, which lasted for >1 month.

Results

Demographics and preoperative symptoms

Twenty-four patients (12 males and 12 females) were recruited in this retrospective review between January 2015 and March 2018. The median (quartile deviation, QD) age of all patients was 30.5(13.8) years. Fifteen patients presented left-side zygomatic fractures, and nine patients presented right-side fractures. Twenty fractures resulted from traffic accidents, two from falls, and two from human altercations. All patients presented swelling, ecchymosis, and facial asymmetry (Figure 4A and 4B). Two patients had preoperative diplopia, and eight suffered from ION anesthesia. The median (QD) clinical follow-up time was 279.5 (48.5) days (Table 1).

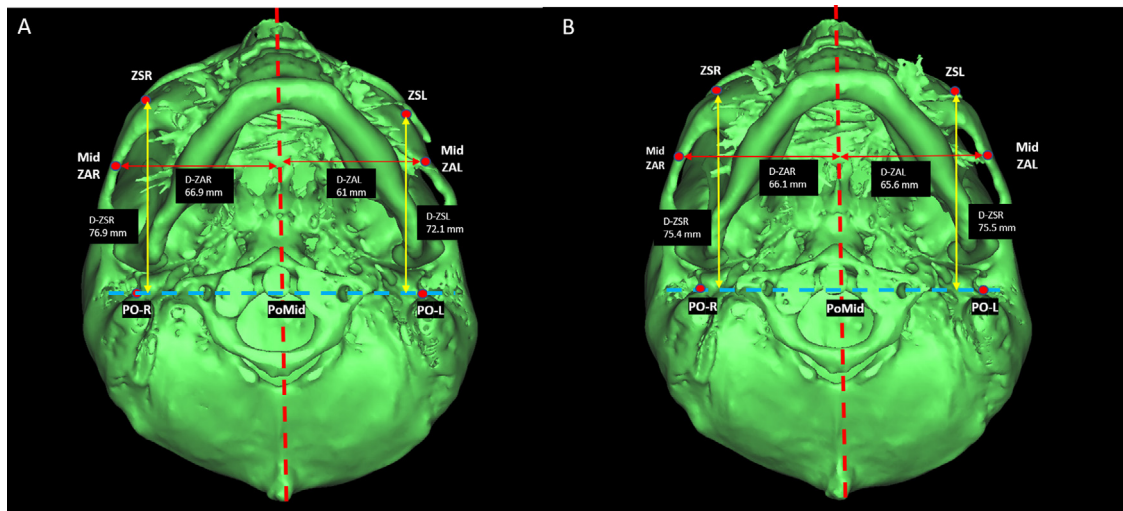


Figure 3 Three-dimensional evaluation. D-ZAR, D-ZAL, D-ZSR, and D-ZSL were measured preoperatively (A) and postoperatively (B). Mid-porion (MP) plane: red dotted line and Porion (P) plane: blue dotted line.

(Porium: PO; Midpoint of porions: PoMid; Zygomatic arch: ZA; Zygomatic surface: ZS; Distance of ZA: D-ZA; Distance of zygomatic surface: D-ZS; R: right; and L: left).



Figure 4 Pre- and postoperative esthetic outcome

A 68-year-old woman was involved in a motorcycle accident with a contusion on the left side of her face. Left zygoma close fracture and malar depression is shown (A and B). After reduction and fixation through computer-assisted navigation system, a better esthetic result was obtained with the symmetry of the bilateral cheek at 10 months follow-up (C and D).

Table 1 Subject demographics.

Age (y/o) (median, (QD))		30.5(13.8)
Sex	Male	12
	Female	12
Trauma mechanism	Traffic accident	20
	Fall	2
	Human altercation	2
Clinical follow-up period (days) (median, (QD))		279.5 (48.5)
Fracture site	Right	9
	Left	15
Preoperative symptoms	Swelling	24
	Ecchymosis	24
	Asymmetry	24
	Diplopia	2
	ION anesthesia	8
Surgical timing (days) (median, (QD))		11.5 (6.3)
Operation duration (min) (median, (QD))		170 (29.5)
Incision sites	Gingivobuccal sulcus	24
	Subciliary	13
	Preauricular	1
Fixation material	Titanium system	19
	Absorbable plates and screws	5
Fixation number	1-point	6
	2-point	9
	3-point	8
	4-point	1
Fixation location	Lateral ZM buttress	23
	Medial ZM buttress	2
	Anterior maxilla	9
	Infraorbital rim	12
	Lateral orbital rim	1
	Zygomatic arch	2
	Zygomaticofrontal suture	2
	Transient ectropion	1
	ION anesthesia	16
Postoperative complications	Diplopia	2
	Enophthalmos	1

QD, quartile deviation; ZM, zygomaticomaxillary; and ION, infraorbital nerve.

Table 2 Two-dimensional pre- and postoperative measurements.

	Preoperative (mm)						Postoperative (mm)						p-value
	ZM	ZF	IO	ZS	ZT	Sum	ZM	ZF	IO	ZS	ZT	Sum	
Mean	4.78	1.78	3.50	3.06	2.55	15.67	1.22	0.40	0.07	0.55	0.50	2.74	<0.001
SD	3.12	1.65	2.53	2.61	2.84	7.22	1.97	0.71	0.26	0.84	1.02	2.50	
Max	13.97	5.16	11.05	9.45	10.19	36.80	7.60	2.47	1.14	3.06	3.46	7.88	
Min	1.07	0.00	0.00	0.00	0.00	4.92	0.00	0.00	0.00	0.00	0.00	0.00	

ZF, Zygomaticofrontal; IO, Inferior orbital rim; ZS, Zygomaticosphenoidal; ZM, Zygomaticomaxillary; ZT, Zygomaticotemporal; SD, standard deviation; Max, maximum; and Min, minimum. The differences of sum between pre- and postoperation reached statistical significance ($p < 0.001$).

2D radiology CT measurements

The mean value of preoperative and postoperative displacement at each zygomatic suture line was shown in [Table 2](#).

The average sum of displacement pre- and postoperatively was 15.67 ± 7.22 vs. 2.74 ± 2.50 , respectively. The differences of sum between pre- and postoperation reached statistical significance ($p < 0.001$) ([Table 2](#)).

Table 3 Three-dimensional pre- and postoperative measurements.

	Preoperative (mm)						Postoperative (mm)						p-value
	D-ZSR	D-ZSL	diff	D-ZAR	D-ZAL	diff	D-ZSR	D-ZSL	diff	D-ZAR	D-ZAL	diff	
Mean	73.43	72.28	4.09	61.86	59.83	4.89	74.51	74.36	0.46	62.92	62.72	0.71	< 0.001
SD	4.91	5.32	2.12	4.89	5.70	2.59	4.64	4.87	0.35	4.08	3.95	0.44	
Max	85	82.1	8.1	71.3	68	10.3	86.1	86.4	1.5	71.3	70.6	1.5	
Min	67.4	64.3	0.5	52.3	43.2	1.2	67.8	67.3	0	54.9	55.1	0.1	

D-ZSR, distance of right zygomatic surface; D-ZSL, distance of left zygomatic surface; D-ZAR, distance of right zygomatic arch; D-ZAL, distance of left zygomatic arch. Diff, difference; SD, standard deviation; Max, maximum; and Min, minimum. The differences between D-ZAL vs. D-ZAR and D-ZSL vs. D-ZSR pre- and postoperative reached a statistical significance ($p < 0.001$).

3D image measurement

Table 3 displays the pre- and postoperative measurements obtained from the 3D CT images. The differences between D-ZAL vs. D-ZAR and D-ZSL vs. D-ZSR reached a statistical significance pre- and postoperatively ($p < 0.001$).

Operation details

All patients underwent surgery within 1 month after injury, with a mean surgical timing of 13.1 ± 7.8 days after trauma event. The average surgery duration was 187 ± 73 min. All reductions were approached through an upper gingivobuccal sulcus incision. Thirteen patients required another subciliary incision, and one patient needed an additional preauricular incision. Nineteen patients underwent fixation with a titanium system and five patients with resorbable plates and screws. One-point surgical fixation was deemed as sufficient stability in six patients, two-point in nine patients, three-point in eight patients, and four-point in one patient. The lateral zygomaticomaxillary buttress was the most common and important site of fixation, because 23 patients were noted to have surgical fixation at that location. Twelve patients underwent infraorbital rim fixation, and nine underwent fixation on anterior maxilla. Additional fixation points were seen on a few patients, such as medial zygomaticomaxillary buttress (two patients), ZA (two patients), zygomaticofrontal suture (two patients), and lateral orbital rim (one patient), depending on each individual's fracture pattern (Table 1).

Seven patients had concomitant orbital floor blowout fractures, and anatomical preformed titanium plate reconstruction was performed simultaneously. As insufficient height of the reconstructed orbital floor was still present in one patient during operation, 1.5-mm-thick porous polyethylene implants (Medpor) were used to reinforce the orbital floor reconstruction.

Postoperative complications

Postoperative complications included transient effects and chronic issues. One patient had transient ectropion, which subsided within 2 weeks. Cheek and paranasal numbness were the most common chronic complications, which were found in 16 patients and disappeared within 6 months (average of 161 days). Two patients presented with diplopia

postoperatively but this symptom was already seen preoperatively. The diplopia subsided within 2 months after ocular rehabilitation. One patient suffered from postoperative enophthalmos and was found to have a significant orbital floor bone defect after reduction of the zygoma, which leads to orbital floor reconstruction at a later stage (Table 1). No one presented postoperative hypoglobus. An example of surgical outcome of facial symmetry improvement was shown pre- and postoperatively (Figure 4).

Discussion

This study aimed to review the surgical outcomes of using the navigation system in patients who suffered from unilateral zygoma fractures by the assessment of facial symmetry quantitatively. The computer-assisted real-time navigation system of virtual surgical planning has been introduced in craniofacial surgery for precision and efficiency during complex operation¹³. It is a reliable approach, which could be applied on customized implant and hardware placement for the correction of craniofacial deformities¹³⁻¹⁴. It provides real-time confirmation for bone reduction, despite limited visibility, implants placement and also safe dissection in acute major facial fractures, including zygoma, frontal sinus, and Le Fort fractures¹⁵. It is also useful in improving the accuracy of facial bone trauma reduction, bone positioning and fixation, orthognathic surgery planning¹⁶ and orbit reconstruction¹¹, reducing surgical difficulty and possibly operation time¹⁷, minimizing intraoperative errors, and improving operative outcomes¹⁶.

The use of the navigation system has been suggested for most zygomatic fractures¹⁸⁻¹⁹, particularly for comminuted fractures, delayed and malunion fractures²⁰, or gunshot wounds. Comminuted ZMC fractures are often unstable and fragmented, making it difficult to reduce and fix adequately. Van Hout et al. retrospectively reviewed their 153 orbitozygomatic fractures and divided the fractures into type A: incomplete, B: tetrapod, and C: Comminuted.⁴ They found that the revision rate is highest in type C comminuted fractures, and 3D virtual planning, intraoperative imaging, and navigation are benefits in revision surgery of type C fracture.

Our data showed that the bony displacement of each suture was decreased to the level of less than 2 mm after operation; the differences of ZA and anterior point that compared with contralateral unaffected side, have been reached less than 1 mm. These revealed satisfactory re-

sults. With the creation of 3D navigation systems, the experiences of virtual surgical planning for reconstruction of post-traumatic zygomatic deformities have developed progressively. Kim et al. use the navigation system in patients with multiple facial fractures and assessed the outcome by superimposing the 3D CT model. They found the quantitative discrepancy was less than 2 mm (1.49 ± 0.27), which was similar to our result²¹. Yang et al. reported that with navigation, the mean total displacement of zygomatic sutures could be significantly decreased to 0.53 mm, particularly the zygomaticosphenoid suture, which showed almost no gap postoperatively²². Bao et al. used 3D the coordinate system and calculated the asymmetry index²³ to confirm the final position after reduction. They found that the facial asymmetry has been significantly improved at zygomatic prominence and ZA in the navigation group when compared with the nonnavigation group⁷. A randomized, controlled trial, which was conducted by Zhang et al. demonstrated that the accuracy of reduction is 0.59 mm in the navigation group and 1.23 mm in the nonnavigation group with significant difference particularly in the intersection of zygomaticomaxillary suture over inferior orbital rim and the intersection of zygomaticotemporal suture.⁶

However, care should be taken that the navigation system has some shortcomings. First, it is time consuming in the initial planning and registration stage. If the plan or registration was not accurate, the surgical results would be compromised. Second, the technical fee and the disposable material may cost a lot in several countries. Third, it creates the additional wound on the scalp for registration, despite the fact that most do not need the coronal incision approach. The last but not the least, for patients who present congenital facial asymmetry, the surgical plan that was designed by mirroring from the contralateral side may reveal good facial symmetry of zygomatic prominence but actually disturbs the judgement of authentic anatomical reduction and causes discrepancy at other landmarks.

The issue concerning whether the navigation reduces surgical time or not remains controversial. Bergeron et al. reported that when using navigation to treat major facial fractures, the surgical time was reduced by 36.1 percent²⁴. However, the study recruited not only zygomatic fractures, but also Le Fort, frontal, and orbital fractures. In addition, isolated fractures or noncomminuted fractures were excluded in this study, and only complex cases were retained. Bly et al. found that navigation shortened the surgical time by ensuring the appropriate implant placement and reclaimed the time spent with the initial setup²⁵ in orbital fractures. Zhang et al. also showed that there is no difference of surgical time between the navigation and nonnavigation group with 183.70 (153 to 229) min vs 179.00 (162 to 235) min, respectively.⁶ They thought that the time spent on the navigation set-up would be more or less compensated by the time saved in faster reduction of zygoma fracture.

The incision sites and numbers have been frequently discussed for zygoma exposure with navigation use. From literature review, one of the main advantages of using the navigation method is that it can potentially avoid bicoronal and subciliary incisions in certain patients, because the complete surgical exposure of all key points would also become optional²⁶. With the navigation system, even without creating additional facial incision, minimally invasive pro-

cedures could reportedly be performed due to the indirect visualization of most parts of the zygoma²². However, we think that navigation plays a role mainly in the confirmation of reduction result not to decrease incision numbers. If the fracture was communicated or unstable after reduction, then multiple incisions are still required for adequate exposure to facilitate reduction and fixation.

Complications of zygoma fracture, when left untreated, usually involve edema, ecchymosis, malar depression, cheek numbness, diplopia, enophthalmos, and hypoglobus; these occur particularly when an orbital floor fracture presents simultaneously. A midfacial buttress system disruption can cause dysfunction, including trismus and malocclusion, which should be treated with surgery²⁷. All patients in our study group were observed to have swelling, ecchymosis, and facial asymmetry followed by ION anesthesia. No trismus or malocclusion was found. The complication rates were similar to the literature²⁸. Conversely, postoperative complications, including inadequate reduction, persistent ION anesthesia, plate and screw exposure, wound infection with soft tissue flap dehiscence²⁹, and persistent diplopia have been reported. Similar to our research findings, ION anesthesia is the most common sequelae²⁸ and usually resolves within 6 months. This symptom results from either the fracture line that runs through the infraorbital foramen or secondary to a blunt trauma that causes neuropraxia after injury. During surgery, the elevation of the buccal maxillary flap also induces postoperative paresthesia³⁰.

In our study, one patient presented postoperative enophthalmos due to the orbital floor blowout fracture, which occurred after the reduction of zygoma laterally. Postoperative enophthalmos after reduction of a pure zygomatic fracture without orbital floor fracture has been noticed. In 2008, Czerwinski et al. described that the reduction of moderate energy zygoma fractures would not produce enophthalmos or create orbital floor defect significantly; however, in displaced zygoma with orbital floor fragments comminution, the reduction maneuver may cause floor blowout that should be explored and addressed³¹. The solution would be to reconstruct the orbital floor with titanium mesh under the guidance of real-time surgical navigation¹¹. There was no major complication found in our study. Comparatively, all patients recovered with a satisfactory final appearance as reported in the literatures^{7, 22}.

The limitations of this study were its retrospective nature and the limited sample size. Research with larger cohorts and longer follow-up periods should be executed. Besides, intraoperative navigation can confirm the bone reduction precisely but cannot predict soft tissue symmetry. Furthermore, a prospective and randomized study should be conducted in the future for high-grade evidence-based medicine regarding the comparison of surgical outcome between the navigation and nonnavigation group. More useful data, including operation time, cost, revision rate, and accuracy between the navigation and nonnavigation group can be obtained through a randomized and controlled comparison study. In addition, the navigation can only tell the point accuracy but not the three-dimensional accuracy. Additional intraoperative imaging or augmented reality system might be the further research topic in the future to improve its three-dimensional precision.

Conclusion

Anatomical reduction of complicated zygomaticomaxillary fractures has been a challenge for craniofacial trauma surgeons. A real-time surgical navigation system is an effective technique for guidance in the reduction of zygomatic fractures. This technology helps improve the accuracy and precision of bony reduction, particularly for unstable, displaced, and comminuted fractures, thereby resulting in better symmetrical zygoma projection.

Declaration of Competing Interest

None.

Funding Source

None.

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